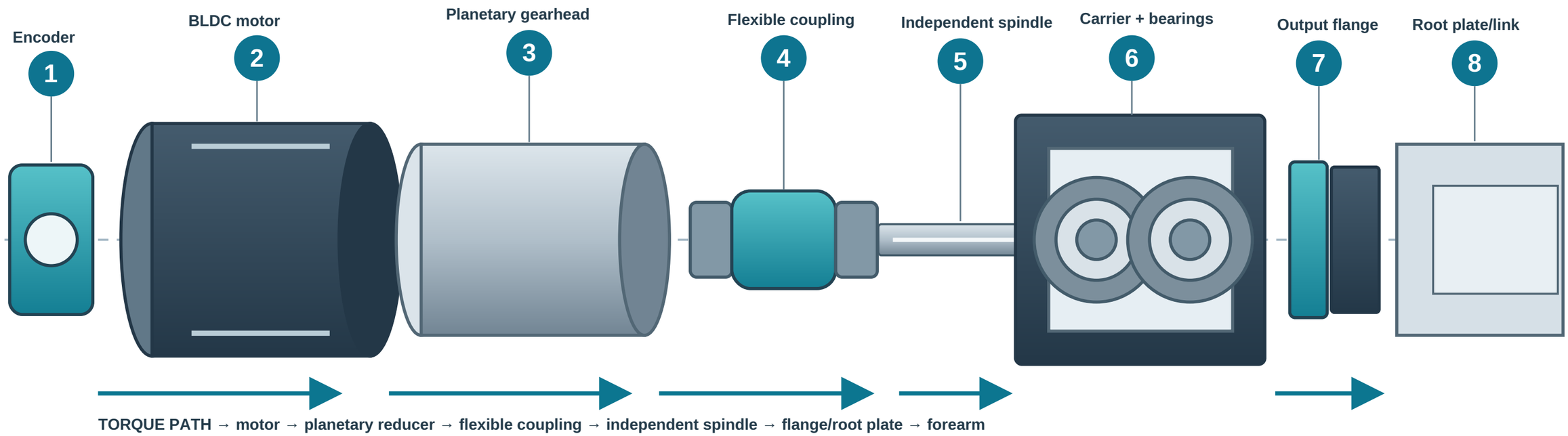
K-EL/01 | Concept Assembly Drawing

KR-MECH-INT-001 • Exploded actuator envelope, numbered items, load-path and support geometry • mm unless noted • Not for manufacture

A. Exploded drive and output support — schematic, not to scale



Catalogue items are envelopes, not vendor CAD. Shaft diameter = 12 mm concept; exact hub bores/keying, retaining features and fits remain open.

B. Numbered item key / proposed BOM

1 Motor-side encoder — maxon ENX16EASY06, 512 CPT candidate

2 BLDC motor — maxon EC-i 52, 48 V, #579165

3 Planetary gearhead — maxon GP 52 C, 91:1, #223094

4 Coupling — KTR ROTEX GS size 19, 98 ShA candidate

5 Independent Ø12 mm output spindle — custom concept

6 Split-yoke carrier + 2 × SKF 6201-2RSH (12×32×10 mm)

7 Output flange — approximate Ø76 mm concept envelope

8 Root mounting plate + illustrative forearm link

C. Concept parameters & load paths

Design basis from brief

1.0 kg at 350 mm; 1.5 kg at 180 mm; output 45 RPM

Static torque 6.0822 N·m; target 12.1644 N·m (assumed 2×)

Motor speed kinematic screen = 45 × 91 = 4095 RPM

Radial load path

Forearm/root plate → spindle → bearing pair → yoke → upper arm

Open question

Confirm dynamic duty, output accuracy/backlash and safe hold on power loss.

D. Notes, trade-offs and release status

- Independent support bearings separate radial loads from reducer torque rating.
- Candidate reducer has ~1° average no-load backlash: ~6.1 mm arc at 350 mm.
- At assumed 40% efficiency, simplified motor-side screen ≈0.334 N·m, 143 W, 3.64 A.
- Supplier to confirm loaded curve, efficiency and the 180 W versus Tω rating point.
- Proposed service: removable cap; bearing retention/access remain unverified. Verify tool reach, cabling and service clearance in detailed CAD.
- No brake selected: externally support when de-energized.

Drawing status: CONCEPT / NOT FOR MANUFACTURE

Source: Kritva Robotics KR-MECH-INT-001, original brief included in release package. All link lengths, carrier geometry and custom parts are illustrative. Confirm exact component configuration and detailed interfaces before any hardware design.